



Day 16: Compute Instances: Choosing CPU vs GPU

A practical guide for AI infrastructure engineers on selecting the right compute architecture for your machine learning workloads.

Why Compute Choice Matters



Foundation

Compute is the essential foundation of AI workload performance, determining processing power and capabilities.



Resource Efficiency

The wrong compute choice leads to wasted time and money—potentially turning a 2-hour job into a 2-day job.



Strategic Decision

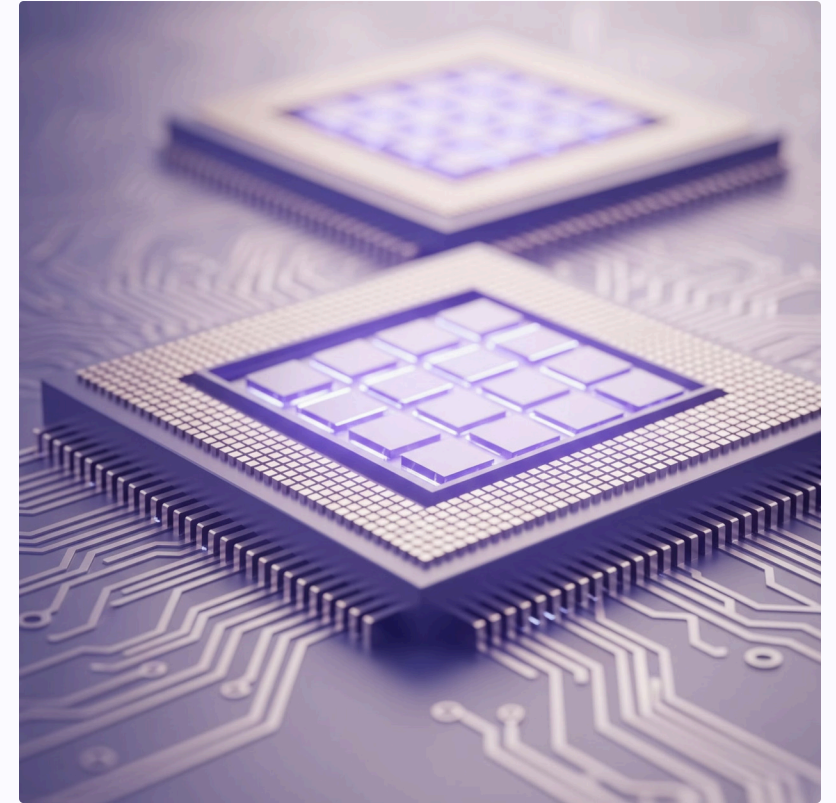
Choosing between CPUs and GPUs depends on workload type, scale, and business requirements.

Making informed compute decisions is a [core skill](#) for AI infrastructure engineers that directly impacts project success.

CPUs – The General Purpose Option

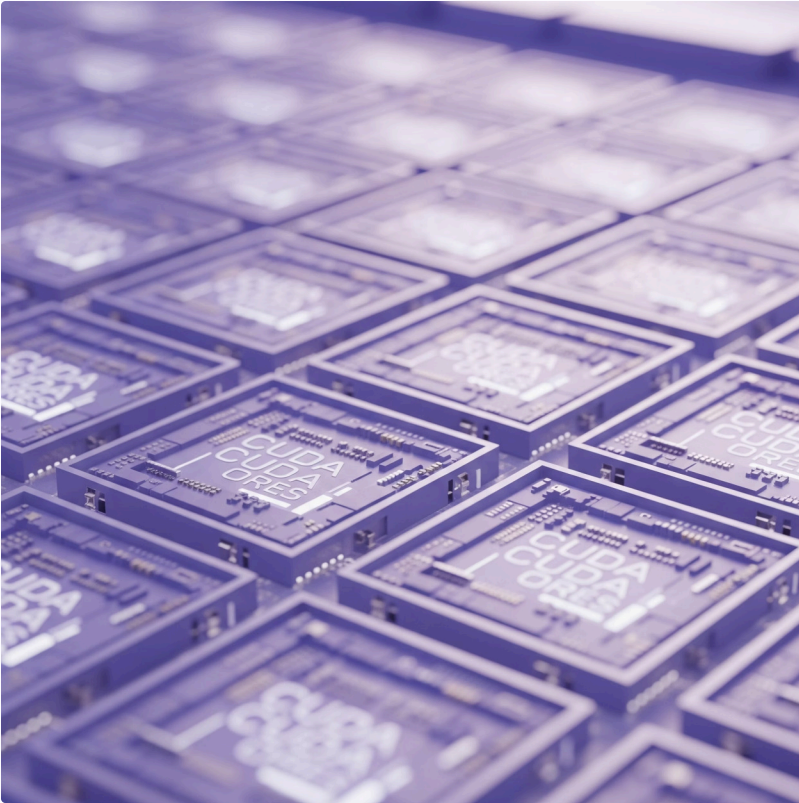
Strengths & Use Cases

- Optimized for **sequential tasks** with complex instruction sets
- Ideal for **data preprocessing & orchestration** workflows
- Cost-effective for lightweight inference jobs
- Excellent for branching logic and non-parallelizable operations
- Widely available across all cloud providers with flexible scaling



Best when parallelism isn't required

GPUs – The Parallel Powerhouses



CUDA cores enable [massive parallelization](#)

Key Capabilities

- Designed for **massive parallel computations** at scale
- Essential for **deep learning training** with large models
- Thousands of cores accelerate matrix operations
- Specialized for vision, NLP, and generative AI workloads
- Available in various tiers: A100, H100, L4 for different needs

Cost Considerations

CPU Economics

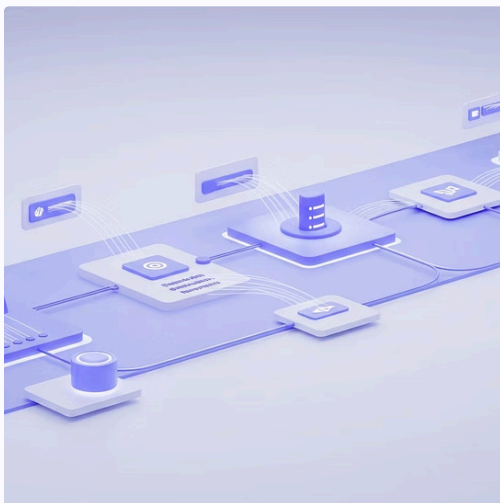
- Lower hourly cost (\$0.05-0.50/hr)
- Higher total cost for AI training
- Good for long-running, low-intensity jobs
- Widely available, rarely capacity-constrained

GPU Economics

- Higher hourly cost (\$1-40/hr)
- Lower total cost for training due to speed
- Spot/preemptible instances cut costs by 70-90%
- Often capacity-constrained (especially H100s)

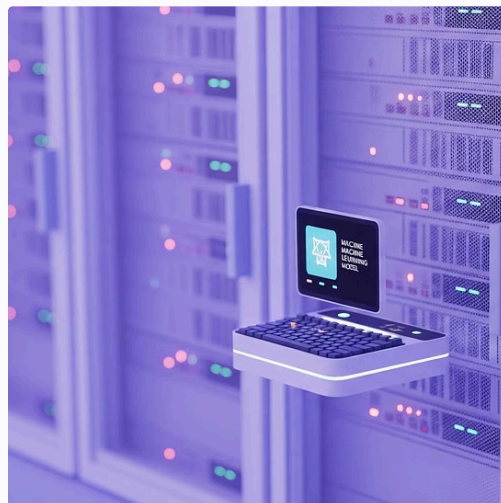
Always [benchmark your specific workload](#) before committing to production infrastructure to balance time-to-train vs. total cost effectively.

When to Use CPUs



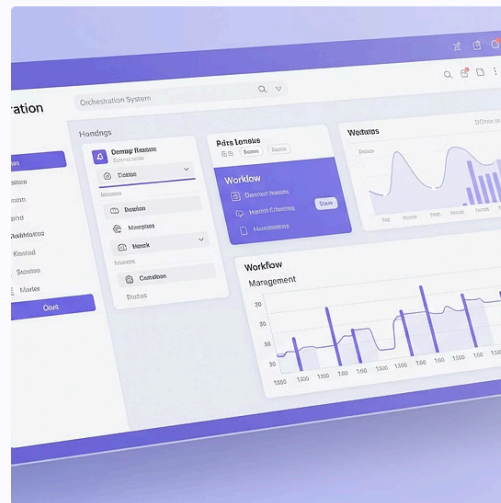
Data Preprocessing

ETL jobs, feature engineering, and data validation workflows



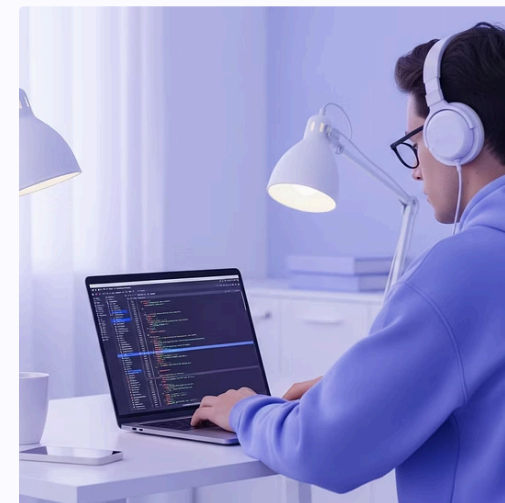
Small ML Models

Serving compact models (scikit-learn, XGBoost) with moderate traffic



Control Plane

Orchestration, scheduling, and monitoring of distributed ML systems



Development

Local testing, prototyping, and development environments

When to Use GPUs

Deep Neural Networks

Training computer vision, NLP, and generative AI models with complex architectures requiring matrix operations at scale

LLM Inference

Serving large language models with high throughput requirements (8-bit quantization still benefits from GPU acceleration)

Distributed Training

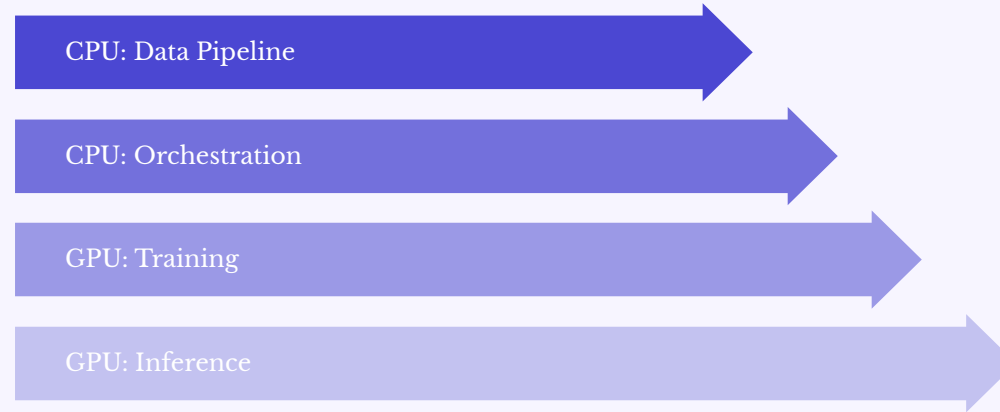
Multi-node training across GPU clusters for foundation models and large-scale experiments requiring data parallelism

Edge AI

Deploying specialized models on edge devices with GPU accelerators (NVIDIA Jetson, L4 cards) for real-time processing



Hybrid CPU + GPU Workloads



CPU Responsibilities

- Data ingestion and preprocessing
- Request routing and load balancing
- Workflow orchestration
- System monitoring and logging

GPU Responsibilities

- Model training computation
- High-throughput inference
- Matrix operations and tensor processing
- Specialized kernels execution

Most production ML systems use [mixed setups](#) to maximize efficiency and balance costs.

Cloud Instance Examples

AWS

- **CPU:** c6i, m6i, r6i series
- **GPU (General):** g5, g6 (NVIDIA T4/L4)
- **GPU (ML):** p4d/p5 (A100/H100)
- **GPU Memory:** Up to 80GB per GPU
- **Pricing:** \$0.05-40/hr depending on type

Google Cloud

- **CPU:** n2, c2, m2 series
- **GPU (General):** g2 (L4)
- **GPU (ML):** a2/a3 (A100/H100)
- **TPU:** v4/v5e for ML-specific workloads
- **Spot Discounts:** Up to 91% off on-demand

Azure

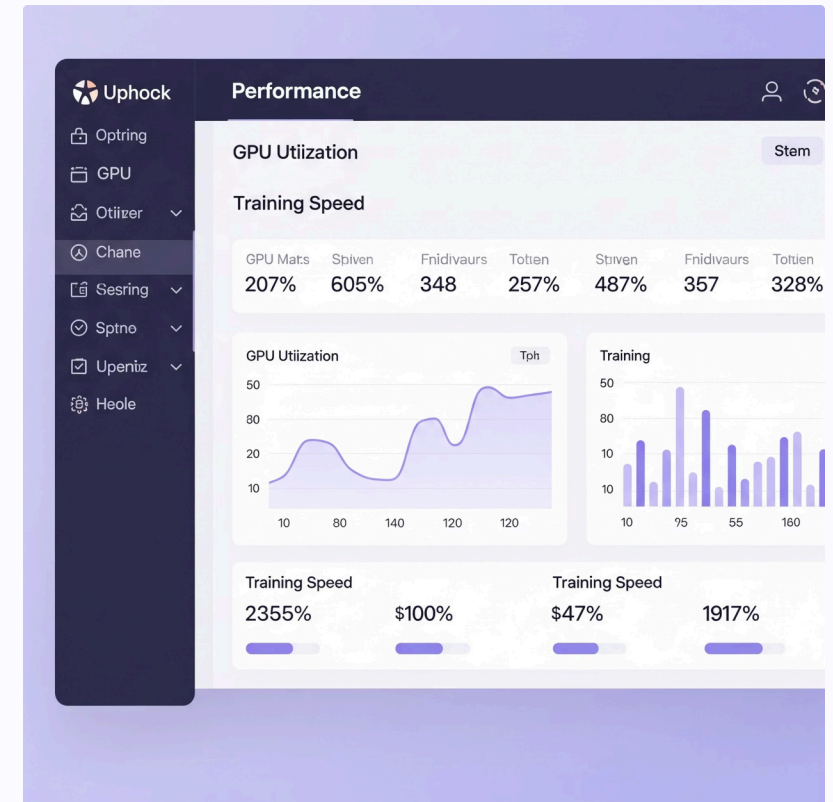
- **CPU:** F-series, Dsv5, Esv5
- **GPU:** NC/ND-series (V100/A100)
- **GPU Clusters:** NDm A100 v4
- **Low Priority:** Significant discounts
- **Reserved Instances:** 1-3 year commitments

Specialized GPU clouds like [CoreWeave](#) and [Lambda Labs](#) often offer better availability for scarce GPU resources.

Benchmarks & Performance Testing

Essential Benchmarking Practices

- **Always test before scaling** with representative workloads
- Use monitoring tools: `time`, `nvidia-smi`, `nsys profile`
- Compare metrics across different instance types:
 - Training time (hours/epochs)
 - Throughput (samples/second)
 - Total cost to completion
- Test with small dataset subsets (~5-10%) before full runs



Benchmark results guide cost-efficient scaling

Key Takeaways

CPUs excel at versatility — ideal for preprocessing, orchestration, and lightweight inference with complex branching logic

Hybrid architectures optimize production systems — most real-world ML platforms use both CPU and GPU resources in concert

GPUs deliver massive parallelism — essential for training deep neural networks and high-throughput inference

Cost optimization requires benchmarking — measure your specific workload performance to make data-driven infrastructure decisions

- ❏ Next Steps: Evaluate your current workloads using the guidelines presented today, and conduct benchmarks to identify opportunities for compute optimization.